UAV Secure Deployment Under Selfish Access Attacks in UAV-Assisted Networks

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***Abstract*—Unmanned Aerial Vehicle (UAV)-assisted access networks emerge as a critical enabler for providing flexible and reliable services across diverse applications. However, due to the limited communication resources of UAVs, users may selfishly increase their transmit power to achieve a higher access success probability in random access scenarios. Existing studies often address selfish access during network operation, which imposes considerable overhead on resource-limited UAVs. This paper proposes a secure UAV deployment method that can defend against potential selfish access attacks during the deployment phase, thereby reducing runtime overhead and improving overall system security and efficiency. We analyze the probability of random access collision, channel gain, and transmit power cost in the UAV-assisted network, and accordingly model the payoffs of different user access strategies and UAV defense strategies. Based on the payoff model, we use the evolutionary game theory to characterize the interactions between the UAV and users. We then derive the effective security margin, which provides a condition for quickly determining whether a deployment point will lead users to evolve into an honest access state. Based on the margin, we define the secure deployment region and provide a proof. Simulation results show that deployment points within the secure deployment region can lead user strategies to converge to the honest state.**

***Index Terms*—UAV-assisted networks, selfish access attack, secure deployment, evolutionary game, random access.**

I. Introduction

In recent years, unmanned aerial vehicle (UAV)-assisted communication systems have attracted increasing attention for supporting seamless coverage, ubiquitous connectivity, and efficient transmission in 5G-A and 6G networks [1]. With high mobility and low deployment cost, UAVs serve as aerial base stations, relays, or access nodes for coverage enhancement [2], [3], communication relaying, and data broadcasting and collection [4], [5]. However, the limited communication or computational resources of UAVs and the openness of wireless channels expose UAV-assisted networks to serious security threats in multi-user random access scenarios [6], [7]. Specifically, simultaneous access attempts may cause collisions and lead to transmission failures [8], while malicious users can increase their transmit power to dominate channel access and improve the access success probability [8]–[10]. Such selfish access attacks exacerbate the unfairness of access competition and degrade honest users' quality of service (QoS) [6], [11].

Several studies have investigated defenses against selfish access. The work [12] addressed selfish access in IEEE 802.11ah networks by placing selfish stations in restricted access window groups with fewer access opportunities. The authors in [13] detected abnormal channel occupation by selfish nodes and mitigated it through cooperative backoff adjustment. Wang *et al.* [6] modeled channel access attacks in low-altitude wireless networks as a Stackelberg game in which the legitimate drone and ground base station adapt to reduce performance degradation. However, these studies mainly defend against selfish behaviors after the UAV starts service, which can impose considerable sensing, computation, and signaling burdens on resource limited UAVs. Therefore, preventing such attacks at the UAV deployment stage before operation is of great value. Since the UAV deployment location determines the air-to-ground channel quality and further affects the attack cost, selecting a proper deployment location can reshape user payoffs such that selfish high-power access attacks become unattractive.

Evaluating UAV deployment requires modeling how user access strategies evolve, for which game theory is widely used in communication networks. The work [14] modeled the transmission strategies of multiple base stations under QoS constraints as a generalized Nash equilibrium problem. The authors in [15], [16] modeled noncooperative spectrum sharing in cognitive MIMO networks, focusing on energy efficiency and robustness to imperfect channel information, respectively. Among these approaches, evolutionary game theory (EGT) is well suited to characterizing strategy evolution because it captures boundedly rational updates driven by payoff differences [17]. The work [18] investigated the access-mode evolution of mobile users in heterogeneous sub-6 GHz/mmWave cellular networks, with the payoff characterized by achievable rate, and handover overhead. In UAV-assisted networks, the authors in [19] used EGT to describe how mobile devices select UAV base stations, where UAV bandwidth, service price, and user association ratio are the main factors affecting the payoff.

Despite the growing interest in access defense and EGT-based access modeling, several challenges remain. First, existing work on defending against selfish high-power access attacks mainly relies on runtime monitoring and adaptive adjustment, which brings continuous considerable overhead to resource-limited UAVs. Second, most studies have not jointly characterized the combined impact of UAV deployment, channel quality, random access collisions, and transmit cost on the payoffs and strategy evolution of the users and UAVs. Third, it is difficult to quantify how user strategies will change under a specific deployment location, such as whether honest users will alter their strategies due to the greater benefits of selfish access.

To tackle these challenges, we develop a UAV secure deployment method for UAV-assisted networks under selfish access attacks. We establish an evolutionary game model to characterize the UAV-users' interactions and analyze the stability of the honest access state. Furthermore, we derive the security margin and the safe UAV deployment region. Our main contributions can be summarized as follows:

- We propose a security UAV deployment method that can defense the selfish access attacks. This method incentives user to access honestly via UAV placement before service operation and reduces the overhead of continuous runtime monitoring.
- We establish payoff models for users and the UAV in UAV-assisted access networks. The models jointly consider the UAV deployment location, channel gain, random access collisions, and transmit power cost to characterize the payoffs of both users and the UAV under different strategies.
- We characterize the interaction process between the UAV and users by the EGT. Then, we derive the security margin to determine whether a deployment point can ensure users' honest access. Furthermore, we define the secure deployment region and prove that users can be stable in the honest access state.
- Numerical results show how UAV deployment positions, user dispersion, access load, and capture threshold affect the safe deployment result. The evolutionary trajectories further verify that the proposed margin predicts the long-term system outcome.

II. System Model and Payoff Formulation

This section formulates the system model and the payoffs that UAV deployment affects. We first define the UAV-assisted networks, and then derive the user access payoff and UAV defense payoff.

A. System Model

As illustrated in Fig. 1, we consider a UAV-assisted uplink random access network, where a UAV serves a cluster of ground users. Due to the limited communication resources of the UAV, multiple users may compete for the shared wireless

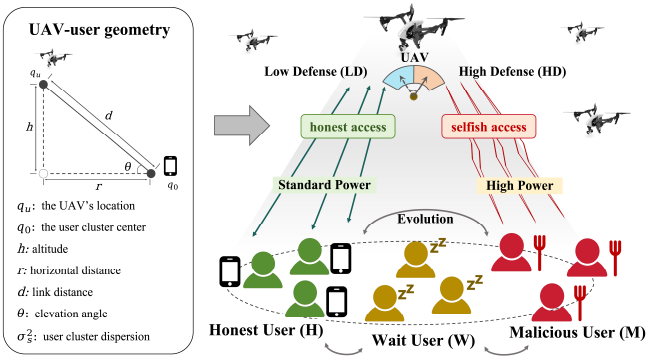


Fig. 1: Illustration of UAV-assisted Access Network Model

channel, which can lead to message collisions and transmission failures. Users may increase their transmission power for selfish access to improve their access success probability, while the UAV can mitigate the negative impact of selfish access by adjusting its deployment location.

The UAV has two defense strategies, $\Omega_A = \{\mathrm{LD}, \mathrm{HD}\}$, where LD denotes low defense and HD denotes high defense with additional mechanisms against selfish high-power access. The UAV's strategy state is described by the probabilities p_{LD} and p_{HD}, where $p_{\mathrm{LD}} + p_{\mathrm{HD}} = 1$, representing the probabilities of adopting low defense and high defense, respectively. In contrast, the ground users form an evolutionary population with three access strategies, $\Omega_U = \{\mathrm{H}, \mathrm{M}, \mathrm{W}\}$, where H denotes honest access with normal transmit power, M denotes selfish access, where users employ higher transmit power to improve their access success probability, and W denotes the waiting state, where users avoid accessing the system. Let ρ_{H}, ρ_{M}, and ρ_{W} represent the proportions of the population adopting honest access, selfish access, and waiting state strategies, respectively, which collectively form the state vector $\boldsymbol{\rho} = [\rho_{\mathrm{H}}, \rho_{\mathrm{M}}, \rho_{\mathrm{W}}]^{\mathrm{T}}$ satisfying the condition $\rho_{\mathrm{H}} + \rho_{\mathrm{M}} + \rho_{\mathrm{W}} = 1$.

The UAV location relative to the user cluster determines the channel condition and the resulting payoffs. We denote the center of the ground user cluster by $\mathbf{q}_0 = [x_0, y_0, 0]^{\mathrm{T}}$ and the UAV location by $\mathbf{q}_u = [x_u, y_u, h]^T$, where h is the UAV altitude. The horizontal distance between the UAV and the user cluster center is $r = \sqrt{(x_u - x_0)^2 + (y_u - y_0)^2}$. We characterize the UAV deployment point by (r, h) and define $d(r,h) = \sqrt{r^2 + h^2}$ and $\theta(r,h) = \arctan(h/r)$.

In the air-to-ground link, the LoS state denotes an unobstructed direct propagation path, while the NLoS state denotes a link blocked by urban obstacles. Following the probabilistic LoS/NLoS air-to-ground channel model in [20], we define the channel gain as

$$\begin{aligned} g(r,h) = {} & p_{\mathrm{LoS}}(\theta(r,h))\beta_{\mathrm{LoS}} d(r,h)^{-\alpha_{\mathrm{LoS}}} \\ & + \left(1 - p_{\mathrm{LoS}}(\theta(r,h))\right)\beta_{\mathrm{NLoS}} d(r,h)^{-\alpha_{\mathrm{NLoS}}}, \end{aligned} \tag{1}$$

where $p_{\mathrm{LoS}}(\theta)$ is the LoS probability, β_{LoS} and β_{NLoS} are channel gain constants, and α_{LoS} and α_{NLoS} are the path-loss exponents of LoS and NLoS links, respectively.

To capture user dispersion, we write a generic user location as $\mathbf{q} = \mathbf{q}_0 + \boldsymbol{\epsilon}$, where $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$, $\mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] = \sigma_s^2\mathbf{I}_2$, and $\mathbf{I}_2$ is the two-dimensional identity matrix. For $\mathbf{q} = [x, y, 0]^T$, its horizontal distance to the UAV projection is $r_{\mathbf{q}} = \sqrt{(x_u - x)^2 + (y_u - y)^2}$.

B. User Access Payoff Model

The UAV provides K random access resources in each access period.

Only honest and selfish users transmit; waiting users remain inactive. Let Λ be the average number of potential access attempts in one period. Under Poisson ALOHA [8], users choose uniformly from K resources, so $\lambda_i = \Lambda\rho_i/K$ for strategy $i \in \{\mathrm{H}, \mathrm{M}\}$, and the probability of no collision is

$$p_0 = \exp\left(-\lambda_{\mathrm{H}} - \lambda_{\mathrm{M}}\right). \tag{2}$$

Let σ^2 denote the additive white Gaussian noise power at the UAV receiver. An honest user sets its transmit power as $P_{\mathrm{H}}(r, h) = P_R/g(r, h)$ to achieve a fixed target received power P_R at the UAV. In contrast, selfish users transmit at higher power levels and may capture the random access resource in the event of a collision. To obtain a capture advantage over an honest packet involved in the collision, the selfish user needs to satisfy

$$\frac{P_{\mathrm{M}} g(r, h)}{P_R + \sigma^2} \geq \tau_C, \tag{3}$$

where P_{M} denotes the selfish user's transmit power and τ_C denotes the capture threshold. Assuming selfish users are rational and do not expend unnecessary energy, the required transmit power to exactly meet this threshold is

$$P_{\mathrm{M}}^{\mathrm{req}}(r, h) = \frac{\tau_C \left(P_R + \sigma^2\right)}{g(r, h)}. \tag{4}$$

Let $\chi_D \in [0, 1]$ denote the probability of successfully defending against high-power capture. The access success probabilities of honest and selfish high-power users are

$$\begin{cases} p_{\mathrm{H}} = p_0, \\ p_{\mathrm{M}} = e^{-\lambda_{\mathrm{M}}} \left[e^{-\lambda_{\mathrm{H}}} + \left(1 - e^{-\lambda_{\mathrm{H}}}\right)\left(1 - p_{\mathrm{HD}}\chi_D\right)\right]. \end{cases} \tag{5}$$

The factor $e^{-\lambda_{\mathrm{M}}}$ accounts for collisions between selfish packets with similar received power, while χ_D reduces their capture advantage over honest packets.

To evaluate user incentives, we use the above access success probabilities and define the access utilities for an honest user and a selfish high-power user, respectively, as

$$V_{\mathrm{H}} = p_{\mathrm{H}} \log\left(1 + \frac{P_R}{\sigma^2}\right), \tag{6}$$

$$V_{\mathrm{M}} = p_{\mathrm{M}} \log\left(1 + \frac{\tau_C(P_R + \sigma^2)}{\sigma^2 + \frac{\Lambda\rho_{\mathrm{H}}}{K} P_R}\right). \tag{7}$$

The denominator in V_{M} accounts for the average same resource honest interference, while (3) defines the capture threshold against a representative collided honest packet.

Under the assumption of a constant received power at the UAV, both V_{H} and V_{M} do not depend on the UAV deployment location. The impact of the deployment location on user payoffs is reflected through the cost of the users' transmit power.

We impose a maximum user transmit power $P_{\max}$ and exclude deployment points where either $P_H > P_{\max}$ or $P_M^{\mathrm{req}} > P_{\max}$ for most users in the cluster.

We model R_U as a fixed subscription-based access charge, so it does not change with UAV deployment or transmit power. Combining this charge with physical energy consumption, the access costs are

$$C_{\mathrm{H}}(r, h) = c_E \frac{P_R}{g(r, h)} + R_U, \tag{8}$$

$$C_{\mathrm{M}}(r, h) = c_E P_{\mathrm{M}}^{\mathrm{req}}(r, h) + R_U, \tag{9}$$

where c_E converts energy consumption into utility cost. Here $C_i(r, h)$ describes a representative user located at the cluster center. To evaluate the population-level incentive, we average the cost over a randomly selected user:

$$\bar{C}_i(r, h, \sigma_s^2) = \mathbb{E}_{\mathbf{q}}(C_i(r_{\mathbf{q}}, h)), \quad i \in \{\mathrm{H}, \mathrm{M}\}. \tag{10}$$

We approximate this expectation by applying a second-order Taylor expansion to $C_i(r_{\mathbf{q}}, h)$ around the cluster center. The zero mean perturbation removes the first order term, while the covariance $\sigma_s^2\mathbf{I}_2$ yields the horizontal Laplacian correction:

$$\bar{C}_i \approx C_i(r, h) + \frac{\sigma_s^2}{2}\Delta_\perp C_i(r, h), \quad i \in \{\mathrm{H}, \mathrm{M}\}, \tag{11}$$

where $\Delta_\perp$ is the two-dimensional horizontal Laplacian with respect to the ground user location.

Combining the aggregate utilities and costs, we define the honest, selfish, and waiting state payoffs as

$$\begin{cases} f_{\mathrm{H}} = V_{\mathrm{H}} - \bar{C}_{\mathrm{H}}, \\ f_{\mathrm{M}} = V_{\mathrm{M}} - \bar{C}_{\mathrm{M}}, \\ f_{\mathrm{W}} = u_{\mathrm{W}}, \end{cases} \tag{12}$$

where u_{W} is the constant baseline payoff from the waiting state and does not depend on the UAV deployment point.

These payoffs reveal a deployment tradeoff: a close UAV lowers the cost of selfish high-power access, whereas a distant UAV raises the cost of honest access and may make waiting more attractive.

C. UAV Defense Payoff Model

For the UAV, Π represents a service payoff from the UAV perspective that counts successful access contribution, the operating loss caused by successful selfish capture, and high defense cost. We denote the successful collision capture factors by

$$\eta_{\mathrm{M}}^{\mathrm{LD}} = e^{-\lambda_{\mathrm{M}}}(1 - e^{-\lambda_{\mathrm{H}}}), \quad \eta_{\mathrm{M}}^{\mathrm{HD}} = \eta_{\mathrm{M}}^{\mathrm{LD}}(1 - \chi_D). \tag{13}$$

Let D_{M} denote the extra operating loss caused by a successful selfish capture with excessive transmit power, mainly including

additional receiver processing and interference mitigation cost at the UAV. We assume $D_{\mathrm{M}} > R_U$ in harmful access cases. The high defense cost increases with the malicious user share as

$$C_D(\rho_{\mathrm{M}}) = C_{D,0} + C_{D,1}\rho_{\mathrm{M}}, \tag{14}$$

where $C_{D,0} > 0$ and $C_{D,1} \geq 0$. Therefore, we express the UAV payoffs under low defense and high defense as

$$\Pi_{\mathrm{LD}} = R_U \rho_{\mathrm{H}} p_0 - \rho_{\mathrm{M}} \eta_{\mathrm{M}}^{\mathrm{LD}} \left(D_{\mathrm{M}} - R_U\right), \tag{15}$$

$$\Pi_{\mathrm{HD}} = R_U \rho_{\mathrm{H}} p_0 - \rho_{\mathrm{M}} \eta_{\mathrm{M}}^{\mathrm{HD}} \left(D_{\mathrm{M}} - R_U\right) - C_D(\rho_{\mathrm{M}}), \tag{16}$$

where malicious users still pay the same access charge R_U, and $D_{\mathrm{M}} - R_U$ is the net operating loss of one successful selfish capture.

Low defense keeps the UAV in a lightweight mode and avoids defense cost, but it leaves the access process more exposed to successful selfish capture. High defense consumes defense resources to suppress such capture, so the UAV favors it only when the avoided operating loss exceeds the defense cost.

III. Evolutionary Game and Deployment Stability Analysis

Based on the payoff models in Section II, we formulate the evolutionary game and derive the deployment stability condition for honest access.

A. Game Definition and Replicator Dynamics

For a given UAV deployment point (r, h), we define the evolutionary game with deployment effects as

$$\mathcal{G}(r,h) = \{\Omega_A, \Omega_U, p_{\mathrm{HD}}, \boldsymbol{\rho}, \Pi_{\mathrm{LD}}, \Pi_{\mathrm{HD}}, f_{\mathrm{H}}, f_{\mathrm{M}}, f_{\mathrm{W}}\}, \tag{17}$$

where Section II constructs all strategy sets, state variables, and payoff functions. This subsection uses these payoffs to build the adaptive dynamics of the UAV and the users.

Replicator dynamics describe how the probability or population share of a strategy changes according to its payoff advantage [17]. A strategy with a payoff higher than the average grows, while a strategy with a lower payoff declines. In this paper, these dynamics provide the adaptive law for updating the UAV defense probability and the users' strategies. Since the UAV chooses between low defense and high defense, its update is driven by the payoff difference

$$\begin{aligned}\Delta_\Pi &= \Pi_{\mathrm{HD}} - \Pi_{\mathrm{LD}} \\ &= \rho_{\mathrm{M}} \left(\eta_{\mathrm{M}}^{\mathrm{LD}} - \eta_{\mathrm{M}}^{\mathrm{HD}}\right)\left(D_{\mathrm{M}} - R_U\right) - C_D(\rho_{\mathrm{M}}) \\ &= \rho_{\mathrm{M}} e^{-\lambda_{\mathrm{M}}}(1 - e^{-\lambda_{\mathrm{H}}})\chi_D \\ &\quad \times \left(D_{\mathrm{M}} - R_U\right) - C_D(\rho_{\mathrm{M}}).\end{aligned} \tag{18}$$

The first term in (18) is the loss avoided by high defense when one selfish user collides with honest users, no other selfish user selects the resource, and the capture is blocked. Subtracting the defense cost gives Δ_Π, the advantage of high defense.

Accordingly, the UAV increases the probability of high defense when $\Delta_\Pi > 0$, and decreases it when $\Delta_\Pi < 0$. The following replicator dynamic describes this adaptive adjustment:

$$\dot{p}_{\mathrm{HD}} = p_{\mathrm{HD}}(1 - p_{\mathrm{HD}})\Delta_\Pi. \tag{19}$$

The users evolve according to the replicator dynamics

$$\dot{\rho}_i = \rho_i(f_i - \bar{f}), \quad i \in \{\mathrm{H}, \mathrm{M}, \mathrm{W}\}, \tag{20}$$

where i denotes the user strategy and $\bar{f} = \rho_{\mathrm{H}} f_{\mathrm{H}} + \rho_{\mathrm{M}} f_{\mathrm{M}} + \rho_{\mathrm{W}} f_{\mathrm{W}}$ is the average user payoff. These dynamics are later linearized around the honest access state to obtain the Jacobian and the stability margins.

B. Honest Access Equilibrium and Security Margins

We focus on the honest access state

$$E_{\mathrm{H}} = (p_{\mathrm{HD}}, \rho_{\mathrm{H}}, \rho_{\mathrm{M}}, \rho_{\mathrm{W}}) = (0, 1, 0, 0), \tag{21}$$

where all users adopt honest access, and the UAV stays in low defense. To examine whether this state can persist, we compare the payoffs of honest access, selfish high-power access, and waiting state after substituting E_{H}. The three equilibrium payoffs are

$$\begin{cases} f_{\mathrm{H}}|_{E_{\mathrm{H}}} = V_{\mathrm{H}}|_{E_{\mathrm{H}}} - \bar{C}_{\mathrm{H}}\big|_{E_{\mathrm{H}}}, \\ f_{\mathrm{M}}|_{E_{\mathrm{H}}} = V_{\mathrm{M}}|_{E_{\mathrm{H}}} - \bar{C}_{\mathrm{M}}\big|_{E_{\mathrm{H}}}, \\ f_{\mathrm{W}}|_{E_{\mathrm{H}}} = u_{\mathrm{W}}. \end{cases} \tag{22}$$

The honest state can be locally stable only when the honest payoff at E_{H} is larger than the selfish high-power payoff and the waiting payoff at E_{H}. Therefore, we define two payoff margins:

$$\begin{aligned}\Delta_{\mathrm{HM}} &= f_{\mathrm{H}}|_{E_{\mathrm{H}}} - f_{\mathrm{M}}|_{E_{\mathrm{H}}} \\ &= \left(V_{\mathrm{H}} - V_{\mathrm{M}} + \bar{C}_{\mathrm{M}} - \bar{C}_{\mathrm{H}}\right)\big|_{E_{\mathrm{H}}}, \\ \Delta_{\mathrm{HW}} &= f_{\mathrm{H}}|_{E_{\mathrm{H}}} - f_{\mathrm{W}}|_{E_{\mathrm{H}}} \\ &= \left(V_{\mathrm{H}} - \bar{C}_{\mathrm{H}} - u_{\mathrm{W}}\right)\big|_{E_{\mathrm{H}}},\end{aligned} \tag{23}$$

where both margins are functions of the UAV deployment point (r, h) and the user dispersion σ_s^2.

Δ_{HM} and Δ_{HW} compare honest access with selfish high-power access and waiting, respectively. A nearby UAV may favor selfish access, whereas a distant UAV raises the honest transmit cost and favors waiting.

Based on the above two margins, we define the effective security margin as

$$S_{\mathrm{eff}}(r, h, \sigma_s^2) = \min\left\{\Delta_{\mathrm{HM}}, \Delta_{\mathrm{HW}}\right\}. \tag{24}$$

This margin selects the weaker of Δ_{HM} and Δ_{HW}. If $S_{\mathrm{eff}} > 0$, honest access yields a higher payoff than both selfish access and waiting state; if $S_{\mathrm{eff}} < 0$, at least one of the strategies has a higher payoff than honest access. Thus S_{eff} converts the two payoff comparisons into one deployment stability indicator.

We then define the safe deployment region as

$$\mathcal{S}_{\mathrm{safe}} = \left\{(r, h) \in \mathbb{R}_+^2 : S_{\mathrm{eff}} > 0\right\}. \tag{25}$$

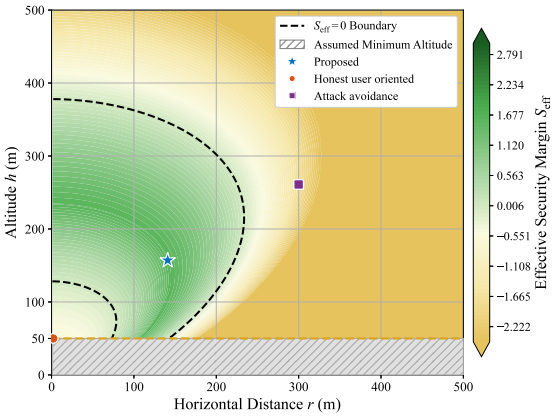


Fig. 2: Spatial distribution of the effective security margin S_{eff} over the (r, h) slice with representative deployment points.

The resulting region excludes deployments that make selfish high-power access profitable and deployments that make honest access too costly to sustain.

C. Stability Guarantee for Safe Deployment Region

Theorem 1: Under the proposed evolutionary game with deployment effects, if $S_{\text{eff}} > 0$ and $C_{D,0} > 0$, then the honest access state $E_{\text{H}} = (0, 1, 0, 0)$ is locally asymptotically stable under the adaptive dynamics of the UAV and users.

Proof: For a fixed feasible deployment, substitute $\rho_{\text{H}} = 1 - \rho_{\text{M}} - \rho_{\text{W}}$ and define $z = (p_{\text{HD}}, \rho_{\text{M}}, \rho_{\text{W}})$. At E_{H}, $z = 0$, $\bar{f} = f_{\text{H}}|_{E_{\text{H}}}$, and $\Delta_{\Pi}|_{E_{\text{H}}} = -C_{D,0}$. In the UAV dynamic (19), all cross derivatives contain $p_{\text{HD}} = 0$. In the user dynamics (20), payoff derivative terms contain $\rho_{\text{M}} = 0$ or $\rho_{\text{W}} = 0$. After these terms vanish, linearization at E_{H} and (23) give the three diagonal entries $-C_{D,0}$, $-\Delta_{\text{HM}}$, and $-\Delta_{\text{HW}}$. Thus, the Jacobian is

$$J(E_{\text{H}}) = \begin{bmatrix} -C_{D,0} & 0 & 0 \\ 0 & -\Delta_{\text{HM}} & 0 \\ 0 & 0 & -\Delta_{\text{HW}} \end{bmatrix}. \tag{26}$$

Hence, perturbations in the three state variables are decoupled to first order near E_{H}. Since $S_{\text{eff}} > 0$ makes both margins positive and $C_{D,0} > 0$, the diagonal matrix $J(E_{\text{H}})$ has three negative eigenvalues and is Hurwitz. Small perturbations therefore decay in the linearized system. The replicator vector field is continuously differentiable near E_{H}, so Lyapunov's indirect theorem establishes local asymptotic stability for the nonlinear dynamics.

■

IV. Numerical Results and Discussion

We conduct simulations to evaluate the secure deployment region and validate the evolutionary behavior under representative deployment points. Unless otherwise stated, the key parameters are $\Lambda = 12$, $K = 50$ (thus $\Lambda/K = 0.24$), $\chi_D = 0.9$, $R_U = 0.1$, $D_{\text{M}} = 3$, $C_{D,0} = 0.15$, $C_{D,1} = 0.1$,

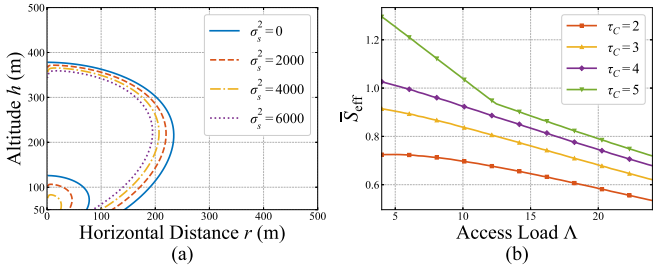


Fig. 3: Sensitivity of deployment viability. (a) Impact of user dispersion σ_s^2 on the safe boundary. (b) Impact of access load Λ on $\overline{S}_{\text{eff}}$ under different capture thresholds τ_C.

and $\sigma_s^2 = 50\text{m}^2$. The horizontal range is $r \in [0, 500]$ m. In the simulations, we set $h \geq 50$ m as an engineering constraint and evaluate only feasible deployment candidates above this altitude. For sensitivity plots, the mean margin over sampled safe candidates is defined as $\overline{S}_{\text{eff}} = \frac{1}{|\mathcal{D}_{\text{safe}}|} \sum_{(r,h)\in\mathcal{D}_{\text{safe}}} S_{\text{eff}}(r, h)$, where $\mathcal{D}_{\text{safe}}$ denotes sampled grid points satisfying $h \geq 50\,\text{m}$, $S_{\text{eff}} > 0$, and the power feasibility constraints.

We assess the spatial distribution of the effective security margin S_{eff} over the (r, h) slice in Fig. 2. The dashed contour marks the boundary $S_{\text{eff}} = 0$, which defines the safe region $\mathcal{S}_{\text{safe}}$. The positive margin region occupies a bounded spatial zone. The figure also marks three representative deployment rules: the proposed rule maximizes S_{eff}, the honest user oriented rule maximizes the honest payoff f_{H}, and the attack avoidance rule selects a deployment with required selfish capture power $P_{\text{M}}^{\text{req}}$ near its maximum feasible value. The latter two points fall outside the safe region because they optimize only one side of the tradeoff, while the proposed point remains inside the safe region by balancing the margins against selfish access and waiting state.

Fig. 3 compares the sensitivity of $\mathcal{S}_{\text{safe}}$ and $\overline{S}_{\text{eff}}$ under different dispersions and access parameters. In Fig. 3(a), increasing σ_s^2 contracts the safe boundary because spatially dispersed users face larger averaged access costs around the cluster. Fig. 3(b) further evaluates $\overline{S}_{\text{eff}}$ as the access load Λ changes. Within the considered load range, a larger Λ reduces the collision-free probability and creates more collision opportunities, so selfish capture becomes more influential than honest access. Therefore, $\overline{S}_{\text{eff}}$ decreases as Λ grows. For the same access load, a larger τ_C requires selfish users to transmit higher power to obtain capture, which raises $\bar{C}_{\text{M}}$ and improves the margin against selfish high-power access.

Finally, Fig. 4 shows the evolution of user and UAV strategy probabilities at the three representative deployments marked in Fig. 2. For the proposed point with $S_{\text{eff}} > 0$, the users converge to honest access and the UAV defense probability decays to zero, which is consistent with the local stability of E_{H}. For the honest user oriented point, selfish high-power access becomes dominant because $\Delta_{\text{HM}} < 0$. For the attack avoidance point, the waiting state becomes dominant because $\Delta_{\text{HW}} < 0$. These trajectories show that S_{eff} is not only a

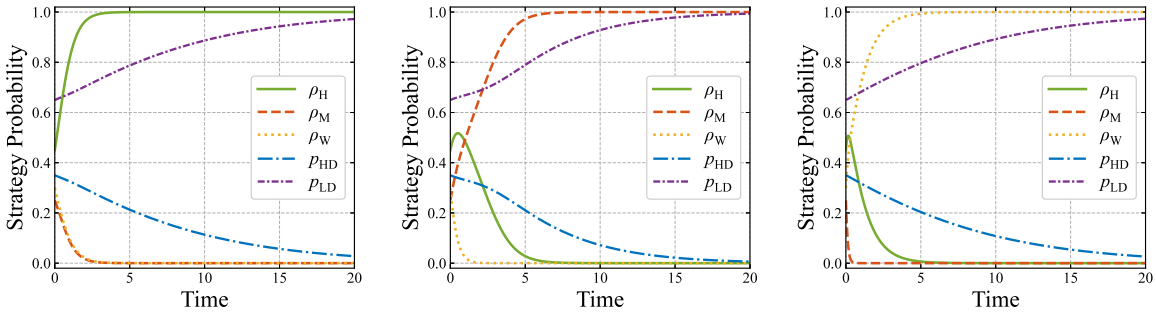


Fig. 4: User and UAV strategy probabilities under different UAV deployments. (a) Proposed deployment with $S_{\mathrm{eff}} = 1.62$. (b) Honest user oriented deployment with $S_{\mathrm{eff}} = -0.59$. (c) Attack avoidance deployment with $S_{\mathrm{eff}} = -1.71$.

static payoff comparison, but also a local stability indicator that follows from the evolutionary game dynamics.

V. Conclusion

This paper investigated secure UAV deployment for UAV-assisted access networks under selfish high-power access attacks. We established a payoff model that links UAV deployment location, channel gain, random access collisions, capture effects, and transmit power cost to the strategic interaction between the UAV and users. Based on the local stability analysis of the honest access equilibrium, we derived an effective security margin and used it to characterize the secure deployment region of the UAV. Numerical results verified the secure deployment region and showed that the proposed margin predicts the long-run evolutionary outcome.

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